

Type: DSC 2-4 (Minor) B.Sc(ECS)-II (Semester III) Course Title: Advanced Python Programming (Paper Code:)		
Credits: Theory – (2) Practical – (1)		
Total Lectures: 30 Hrs.		Contact Hrs. (L) : 2
University Evaluation: 30 Marks		Internal Evaluation: 20 Marks
Course Outcomes:		
1.	To know advanced conditional and looping statements in Python.	
2.	To understand the Regular Expression for pattern matching.	
3.	To understand the parsing of JSON, CSV, Excel files.	
Unit	Content	Lect.
I	Advanced conditional and looping statements- List comprehension, Dictionary comprehension, Use of global, local and nonlocal in functions, Lambda/ Anonymous functions, Higher order functions, Type hinting, filter (), map (), reduce () functions, Recursive functions, Iterables, Iterators and Generators, itertools functions- Infinite iterators- count(), cycle(), repeat(), Iterators terminating on the shortest input sequence- accumulate(), batched(), chain(), chain.from_iterable(), compress(), dropwhile(), filterfalse(), groupby(), islice(), pairwise(), starmap(), takewhile(), tee(), zip_longest() , Combinatoric iterators-product(), permutations(), combinations(), combinations_with_replacement(), Closures and Decorators, @property decorator, Magic methods, *args and **kwargs, Memory management, Garbage collection, Shallow copy Vs Deep copy. Regular expression- Introduction to regular expression, Advantages and Operations, Specify pattern using RegEx-Meta characters- Square brackets [], Period '.', Dollar '\$,' Star '*', Plus '+', Question Mark '?', Curly braces '{}', Alternation ' ', Group '()', Backslash '\', Sequence characters in RegEx- '\A', '\b', '\B', '\d', '\s', '\S', '\w', '\W', '\Z'. functions and constants in re module- findall(), split(), sub(), subn(), search(). Match object-match.group(), match.start(), match.end() and match.span(), match.re and match.string, match...case Statement, Raw string using r prefix.	18
II	Advanced file handling- Use of with statement, Serialization and Deserialization, working with json file- json file, json module, convert dict to json string, convert dict to json file, convert json string to dict, convert json file to dict. Working with CSV Files- CSV File, csv module, reading csv file with csv, Reading CSV	12

	files into a dictionary with csv, Writing CSV files with csv, Writing CSV file from a dictionary with csv. Working with excel file- Basic excel terminology, installing openpyxl, Reading excel spreadsheets, Creating a Simple Spreadsheet, Basic Spreadsheet Operations- Adding and Updating Cell Values, Managing Rows and Columns- insert row, delete row, insert column, delete column. Managing Sheets, Freezing Rows and Columns, Adding Filters, Adding Formulas, Adding Styles, Conditional Formatting, Adding Images- Install pillow module, add image into spreadsheet. Adding Charts like Barchart, Linechart, Pie-chart into spreadsheet.	
Reference Books:		
1.	Programming through Python, M. T. Savaliya, R. K. Maurya, G. M. Magar, STAREDU Solutions	
2.	Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools, Davy Cielen, Arno D.B. Meysman, et al.	
3.	Python Programming: An Introduction To Computer Science, Fourth Edition Grayscale Indian Edition, Zohn Zelle	